Visualisation Design Book

Title Page

Project Title: Hospitalised Injuries from Road Crashes in Australia: Trends, Regional Differences and First Nations Peoples

Team Name: Group 2

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Introduction

1.1 Background and Motivation

Road traffic crashes are an important public health concern because their consequences extend beyond fatalities to include injuries that require hospital admission. These injuries can affect individuals and families, create demands on healthcare services, and contribute to the wider social and economic burden of road trauma. Understanding patterns in hospitalised road crash injuries is therefore important for examining the broader impact of road crashes and supporting informed discussions about road safety.

The Bureau of Infrastructure and Transport Research Economics (BITRE) publishes statistics on hospitalised injuries resulting from road traffic crashes in Australia. Its September 2023 release provides data covering calendar years 2011 to 2021, including national summary statistics, state and territory information, and separate data relating to First Nations Peoples. The data were produced by the Road Safety Data Hub using information sourced from the Australian Institute of Health and Welfare (AIHW) (Bureau of Infrastructure and Transport Research Economics [BITRE], 2023a). These resources provide a foundation for investigating changes over time and differences across selected geographical and population groups.

Although statistical tables contain valuable information, it can be difficult for users to identify long-term patterns, compare regions, and understand differences between population groups by examining figures alone. An interactive data visualisation can present these statistics in a more accessible and structured form. Appropriate charts, annotations, and user interactions can help users recognise trends, compare selected categories, and explore the evidence supporting the project's findings.

The target audience includes members of the public interested in road safety, students and researchers exploring transport-related injuries, and policy or public health stakeholders seeking an overview of hospitalised road crash injuries in Australia. These users may have different levels of statistical knowledge; therefore, the website should communicate its findings clearly without requiring advanced analytical skills.

The main problem addressed by this project is how to transform published statistical data into a clear, accessible, and evidence-based visual story. Rather than displaying all available information without direction, the project will organise selected findings around specific research questions. Users will be able to examine changes in hospitalised injury counts over time, compare geographical patterns, and explore the information available for First Nations Peoples.

The visualisation will also acknowledge that the dataset has limitations. In particular, hospitalised injury counts do not represent every road crash or every road-related injury. Differences in absolute counts should not automatically be interpreted as differences in individual risk, because population size, data coverage, and reporting practices may influence the results. By communicating these limitations alongside the visual evidence, the project aims to encourage careful and informed interpretation of road safety statistics.

1.2 Visualisation Purpose

The purpose of this project is to design and implement an interactive website that communicates meaningful patterns in hospitalised injuries resulting from road traffic crashes in Australia between 2011 and 2021. The website will use D3.js to create interactive data visualisations that support both explanatory communication and user-led exploration.

The project will focus on a structured data story rather than an unrestricted dashboard containing every available variable. The visualisations will be selected according to their relevance to the research questions, the characteristics of the data, and the needs of the intended audience. The final design will aim to help users understand the evidence, identify important patterns, and recognise the limitations of the available statistics.

The project will investigate the following preliminary research questions:

Research Question 1: How did the number of hospitalised injuries resulting from road traffic crashes in Australia change between 2011 and 2021?

Research Question 2: How did hospitalised injury counts and their trends differ across Australian states and territories during this period?

Research Question 3: What patterns can be observed in the hospitalised injury counts reported for First Nations Peoples between 2011 and 2021?

These questions will guide the selection of variables, data processing procedures, exploratory analysis, and visualisation designs. The questions and their scope will be refined if necessary after examining the available data and confirming the definitions in the accompanying data dictionary.

To answer these questions, the website will present temporal trends and selected comparisons using appropriate chart types. A line chart may be used to communicate changes over time, while a bar chart or another suitable comparison chart may be used to compare geographical categories. A separate visualisation may be developed for the First Nations Peoples data, depending on the variables and level of detail available. Interactive features, such as selecting a year, filtering a geographical category, or viewing detailed values through tooltips, will be included where they help users interpret the evidence.

The expected benefits of the completed visualisation are as follows:

1. **Improved understanding:** Presenting statistical information visually will help users identify changes and patterns more easily than relying only on tables.
2. **Clearer comparisons:** Users will be able to examine differences across selected states and territories and explore how those differences change over time.
3. **Evidence-based discussion:** The visualisation will provide a clear basis for discussing hospitalised road crash injuries and the implications of observed patterns.
4. **Greater accessibility:** Clear labels, readable text, appropriate colour choices, and explanatory annotations will help users with different levels of data literacy understand the findings.
5. **More responsible interpretation:** Explicit explanations of data coverage and limitations will reduce the risk of misleading comparisons or unsupported conclusions.

The project is intended to support understanding and discussion rather than establish the causes of road crash injuries or evaluate the effectiveness of particular road safety policies. As the primary statistics represent hospitalised injury counts, the findings will be interpreted descriptively unless suitable additional data are identified and analysed. Any conclusions will be based on the processed dataset and will acknowledge relevant methodological limitations.

Data

2.1 Data Source and Governance

2.1.1 Data Sources

The primary data source for this project is the *Hospitalised Injury* statistical release published by the Bureau of Infrastructure and Transport Research Economics (BITRE). The September 2023 release contains summary series of hospitalised injuries resulting from road traffic crashes for calendar years 2011 to 2021. The data were produced by the Road Safety Data Hub using information sourced from the Australian Institute of Health and Welfare (AIHW), and the published Excel resource also includes raw data (BITRE, 2023a).

The official release provides the following resources relevant to this project:

1. **National hospitalised injury data:** Provides national summary statistics and supporting data for examining changes over time.
2. **State and territory hospitalised injury data:** Provides geographical breakdowns for investigating differences and trends across Australian jurisdictions.
3. **First Nations Peoples hospitalised injury data:** Provides a separate resource for examining the reported hospitalised injury statistics relating to First Nations Peoples.
4. **Data dictionary:** Documents the definitions, scope, and methodological information required to interpret the published data (BITRE, 2023b).

The official data resources are available from the following links:

- BITRE Hospitalised Injury data and Excel files: https://www.bitre.gov.au/resource/road-safety/hospitalised-injury
- Hospitalised Injury Data Dictionary: https://www.bitre.gov.au/sites/default/files/documents/hospitalised-injury-data-dictionary-september-2023.pdf

The accompanying data dictionary is particularly important because the meaning of each variable, the inclusion criteria, and the definition of a hospitalised injury determine how the data should be processed and interpreted. The project will consult this documentation before finalising its data transformations and analytical conclusions.

2.1.2 Data Summary

Table 1 summarises the main characteristics of the selected data source.

Data characteristic	Description
Dataset title	Hospitalised Injury
Publisher	Bureau of Infrastructure and Transport Research Economics (BITRE)

Data characteristic	Description
Data source organisation	Australian Institute of Health and Welfare (AIHW), through the Road Safety Data Hub
Publication date	22 September 2023
Temporal coverage	Calendar years 2011–2021
Geographic coverage	Australia, with separate state and territory data available
Population information	Includes a separate data resource relating to First Nations Peoples
Main measure	Hospitalised injuries resulting from road traffic crashes
Data format	Excel spreadsheets, accompanied by a PDF data dictionary
Access	Publicly available through the official BITRE website
Update frequency	Not assumed to be regular; the publication date and coverage of the selected release will be stated
Number of records	To be confirmed from the specific sheets and tables selected for analysis
Key attributes	To be confirmed from the selected files and data dictionary; expected analytical dimensions include year, geography, and hospitalised injury counts where provided

Table 1. Summary of the BITRE Hospitalised Injury dataset

Note. The exact number of records and the final list of attributes will be documented after the team identifies the relevant worksheets, inspects the raw data, and completes the initial data assessment.

2.1.3 Data Collection and Scope

The published hospitalised injury statistics are based on national hospital separations data obtained from the AIHW National Hospital Morbidity Database. In Australian hospital statistics, a separation refers to an episode of admitted-patient care, such as discharge, transfer, or death. The data dictionary provides further details about the data source, scope, and inclusion criteria (BITRE, 2023b).

For this project, a hospitalised injury refers to a road traffic crash injury that results in a confirmed hospital admission, excluding deaths occurring in hospital from the published hospitalised injury measure. Consequently, the dataset does not represent all road crashes, all people injured in road crashes, or the full range of injuries treated without admission.

The project will use the official documentation to determine which variables and records are suitable for each research question. The national data will support the analysis of overall temporal trends, the state and territory data will support geographical comparisons, and the First Nations Peoples resource will be examined for the third research question, subject to the definitions and coverage of the published data.

2.1.4 Data Quality Assessment

The team will assess data quality before developing the final visualisations. The assessment will consider the following areas:

- **Completeness:** Identify missing values in the variables selected for analysis and determine whether the missing information affects the interpretation of the results.
- **Consistency:** Check that years, geographical labels, category names, and numerical values are represented consistently across the selected files.
- **Validity:** Confirm that data types are appropriate, years fall within the intended study period, and injury counts are valid numerical values.
- **Duplicates:** Inspect the selected tables for duplicate records, distinguishing genuine, repeated observations from accidental duplication before removing anything.
- **Comparability:** Review the data dictionary and relevant methodological notes for changes in definitions, admission criteria, or data coverage that could affect comparisons across years or jurisdictions.
- **Traceability:** Retain the original source files and document any filtering, cleaning, aggregation, or transformation performed during processing.

The results of these checks will be documented in Section 2.2, together with the KNIME workflow and explanations of the decisions made. Any unresolved data quality issues will be reported rather than silently excluded from the analysis.

2.1.5 Data Governance, Privacy, and Ethical Considerations

The project will use the publicly available statistical resources published by BITRE. The source, publication date, coverage period, and relevant methodological documentation will be identified in the Design Book and on the final website. Before redistributing the source data or derived files, the team will check the applicable access conditions, licensing terms, and attribution requirements.

The project will use the published data for statistical analysis and public-interest communication. The team will assess whether the selected files contain any information that could identify individuals and will avoid exposing personal or sensitive information. The analysis and website will focus on the aggregated statistical information appropriate to the research questions.

Particular care is required when presenting the First Nations Peoples data. The visualisation will describe the statistics within the scope defined by BITRE and will avoid unsupported generalisations about First Nations communities. Differences in reported counts will not be presented as evidence of individual behaviour or as proof of a particular cause. Where the data do not support a conclusion, the limitation will be made clear.

The team will also avoid misleading visual encodings. Chart axes, scales, labels, and annotations will be selected to represent the data accurately. Comparisons based on absolute counts will be explicitly identified as count comparisons rather than measures of relative risk. These practices will support responsible communication and help users understand both the value and the limitations of the published information.

2.1.6 Relationship Between the Data and Research Questions

The selected data resources support the project's research questions in different ways.

Research Question 1: National trends over time

The national hospitalised injury data provide the temporal coverage needed to examine how reported counts changed between 2011 and 2021. A time-series visualisation will be used to communicate the direction and scale of changes across the study period. The interpretation will remain descriptive and will not assume that an observed increase or decrease identifies its cause.

Research Question 2: Differences across states and territories

The state and territory data provide a basis for comparing reported hospitalised injury counts across geographical categories and examining how those counts change over time. Appropriate visualisations will be selected after reviewing the structure of the data. Because the figures represent absolute counts, larger counts will not automatically be interpreted as indicating a higher individual risk. Differences in population size and data coverage will be considered when explaining the comparisons.

Research Question 3: First Nations Peoples data

The separate First Nations Peoples resource provides an opportunity to examine the hospitalised injury statistics reported for this population during the available period. The team will verify the definitions, categories, and coverage in the data dictionary and spreadsheet before deciding on the final comparison. Any findings will be described within the limits of the published statistics, without making unsupported causal claims or direct comparisons that the data cannot justify.

Overall, the selected data sources provide a foundation for investigating temporal trends, geographical differences, and the reported statistics for First Nations Peoples. The final analysis will depend on the availability and suitability of the relevant variables. Any limitations identified during data processing and exploration will be incorporated into the visualisation design and the interpretation of the findings.

2.2 Data Processing

Data preparation was completed in KNIME using the workflow Group 2_Road_Injury_Processing. Three separate processing branches were created to prepare national, state and territory, and First Nations hospitalisation data for 2011–2021 (BITRE, 2023a). The resulting CSV files support the project's exploratory analysis and planned website visualisations. The datasets were kept separate because they describe overlapping populations and should not be added together.

National hospitalisation data

The raw worksheet from hospitalisation_injury_publication_sep2023.xlsx was imported using an Excel Reader node. After empty rows were skipped, the imported table contained 117,919 rows and 13 columns. A Row Filter retained records where Cause of injury was Traffic, leaving 86,140 rows relevant to road traffic crashes.

A GroupBy node summed Hospitalisations by Calendar year to produce annual national totals. The existing Hospitalisations field was used without subtracting deaths again. The output columns were renamed to Year and Hospitalisations, sorted by year in ascending order, and exported as national_annual_hospitalisations.csv. This file contains 11 rows, with one record for each year.

State and territory data

The raw worksheet from state-and-territory-hospitalisation-injury-publication-2023.xlsx contains several tables arranged alongside one another. The range A2:F178 was selected to import only the relevant state and territory totals table, producing 176 rows and six columns.

A GroupBy node grouped records by calendar year and state or territory, then summed count of cases excluding died in hospitals within 30 days. This combined the January–June and July–December counts into annual totals. The output columns were renamed to Year, State, and Hospitalisations, then sorted by year and state. The resulting state_annual_hospitalisations.csv contains 88 rows, representing 11 years across eight states and territories.

First Nations data

Initial inspection of the raw worksheet in First-Nations-hospitalised-injuries-2023.xlsx identified four unpublished hospitalisation values marked n.p. in the 2014 age-group breakdown (BITRE, 2023a). These entries caused the hospitalisation column to be imported as text. Replacing them with zero or excluding them from aggregation would understate the annual totals.

To avoid this problem, the published annual totals in T1!A3:C14 were used. This range contains 11 years of hospitalisation counts for First Nations peoples and Non-Indigenous Australians. An Unpivot node converted the two population columns into rows while retaining the calendar year. A Column Filter removed the generated RowIDs column, and the remaining columns were renamed to PopulationGroup, Hospitalisations, and Year.

The records were sorted by year and population group, then exported as first_nations_annual_hospitalisations.csv. This file contains 22 rows, representing two population groups for each of the 11 years.

Output attributes and data types

Attribute	KNIME data type	Description
Year	Integer	Calendar year, from 2011 to 2021
Hospitalisations	Integer	Number of hospitalised road crash injuries
State	String	State or territory category; used in the state dataset
PopulationGroup	String	First Nations peoples or Non-Indigenous Australians; used in the First Nations dataset

Only the attributes required for the current project questions were retained in the exported files. Bed days and detailed demographic breakdowns were not included in these annual summaries. No joins or numerical normalisation were applied because the current questions concern annual hospitalisation counts. Population rates were not calculated.

Validation and data limitations

The three exported CSV files were checked for missing values, duplicate records, expected row counts, and sorting order. No missing values or duplicate records were found in the exported datasets. National totals matched the published annual totals, state and territory values matched the corresponding half-year source counts summed annually, and First Nations values matched the published T1 totals.

Differences between the datasets were also investigated. The eight state and territory totals do not sum to the national total because the national figure includes cases with unrecorded or migratory/offshore residence. Similarly, the two population groups do not sum to the national total because cases with unstated or inadequately described First Nations status are excluded from those groups (BITRE, 2023a).

The source notes identify comparability breaks for Victoria from 2012 and New South Wales from June 2017 (BITRE, 2023b). These limitations must be considered when interpreting trends. The exported values are counts rather than population-adjusted rates, so differences in counts alone should not be interpreted as differences in individual injury risk.

Workflow documentation and reproducibility

Annotations were added to each workflow branch to explain its source, processing steps, output file, and relevant limitations. Reader and writer nodes use paths relative to the workflow's data folder. The exported Group 2_Road_Injury_Processing.knwf includes the three original Excel workbooks, the data dictionary PDF, and all three processed CSV files. All 17 nodes were saved in their executed state.

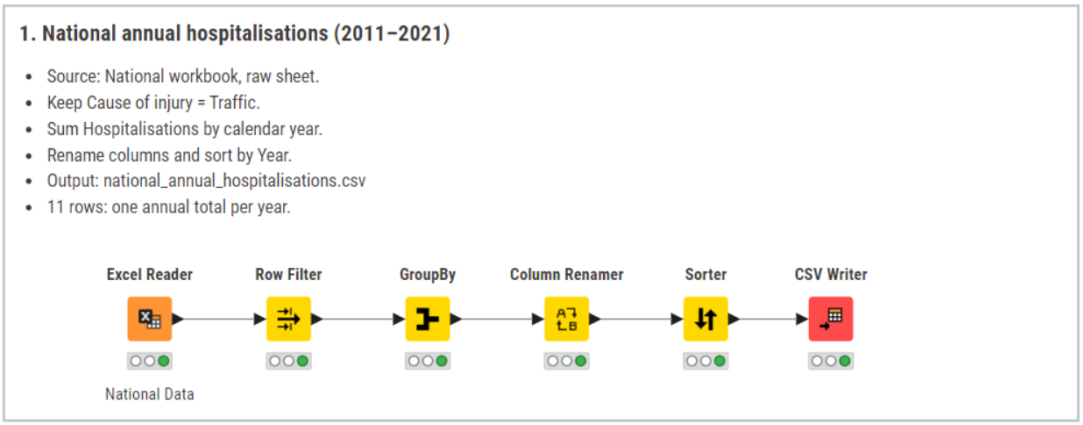


Figure 1. KNIME processing branches for national annual hospitalisation data.

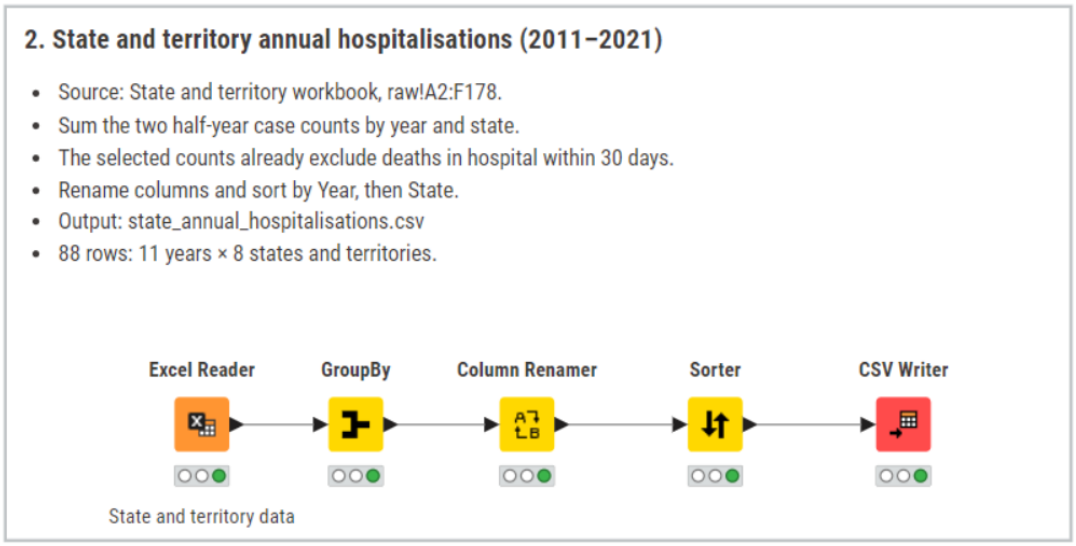


Figure 2. KNIME processing branches for state/territory annual hospitalisation data.

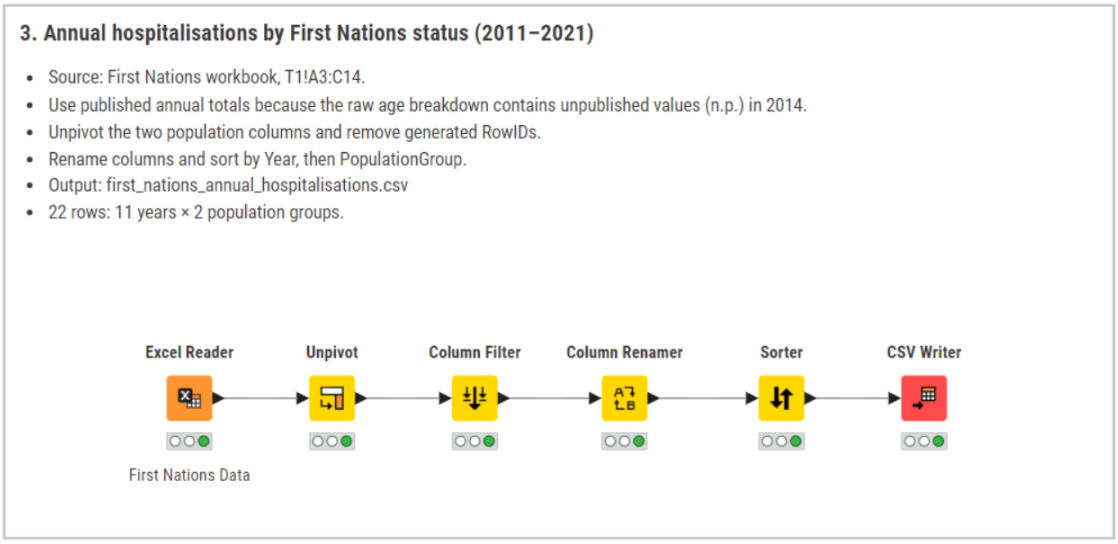


Figure 3. KNIME processing branch for annual hospitalisations by First Nations status.

2.3 Data Exploration

2.3.1

The BITRE file that contains nationwide data about Australia's annual hospitalised injuries consist of summary tables labelled as T1 – T4 with a sheet containing "raw data" about 118,000 rows. The sheet possesses attributes such as Calendar Year, Month, ABS Remoteness area, Cause of Injury, Age Group, Sex, Road User, Counterparty, Count of cases, Bed days, Died_cases, Died_bed_days and Hospitalizations (BITRE, 2023a). The summary tables on the other hand provide us with data regarding the total number of road users, age group, sex and remoteness area.

2.3.2 Initial Findings

Firstly, Table 1 data that represents the Annual Hospitalizations shows a major jump in the number of hospitalised individuals from 2011 to 2019 where it reached a peak of 39,866. From 2019 to 2020, there was a dip of 5% where it fell to 37,966 while it rose to 39,505 in 2021. Although the results were going up and down, 2021 shows that the count was still about 16% higher than 2011.

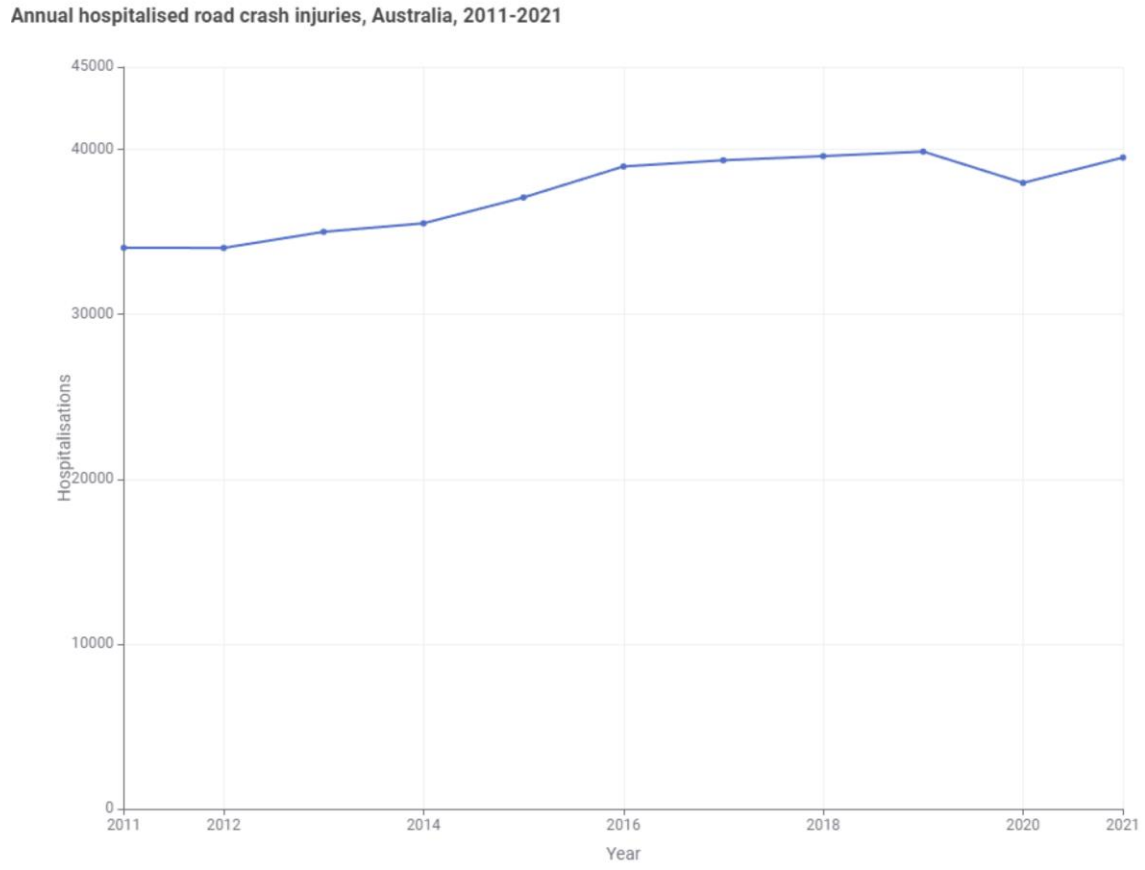


Figure 4. Annual Hospitalized Road crash injuries, Australia, 2011-2021

Secondly, Table 2 represents the data on Annual Hospitalizations by road users. As there are many attributes that have tremendous amounts of data, one that stands out would be the Pedal Cyclist section. Pedal Cyclist individuals had the largest jump which rose from 5,391 in 2011 to 8,163 in 2021. On the other hand, Pedestrians fell 15.3% from 2,754 to 2,334 over the same period. However, the largest group with hospitalizations in 2021 was Car Drivers, recording 13,327 followed by Motorcyclists with a total of 8,883.

Thirdly, Table 3 represents the data on Annual Hospitalizations by age group. From this table, we can make an analysis on how seniors/older adults are leading with ages from 65 to 74 rose from 1,968 to 3,018 and ages 75 and above rose from 1,906 to 2,458 during 2011 to 2021. In addition, data from ages 0-7 and 17-25 was interesting due to its fluctuation where 2021's hospitalization was less compared to 2011 itself.

Fourthly, the State and Territory workbook (excel file) shows annual hospitalization for each jurisdiction. (Refer to Figure 2). During the year 2021, Victoria had the highest count amounting to 11,130 followed by Queensland with 10,447 and New South Wales with 9,003. Between 2011 and 2021, Queensland was recorded as the largest increase, which was 6,322 to 10,447. In addition, Northern Territory and Tasmania did rise but on a much lower scale compared to New South Wales which decreased from 10,618 to 9,003.

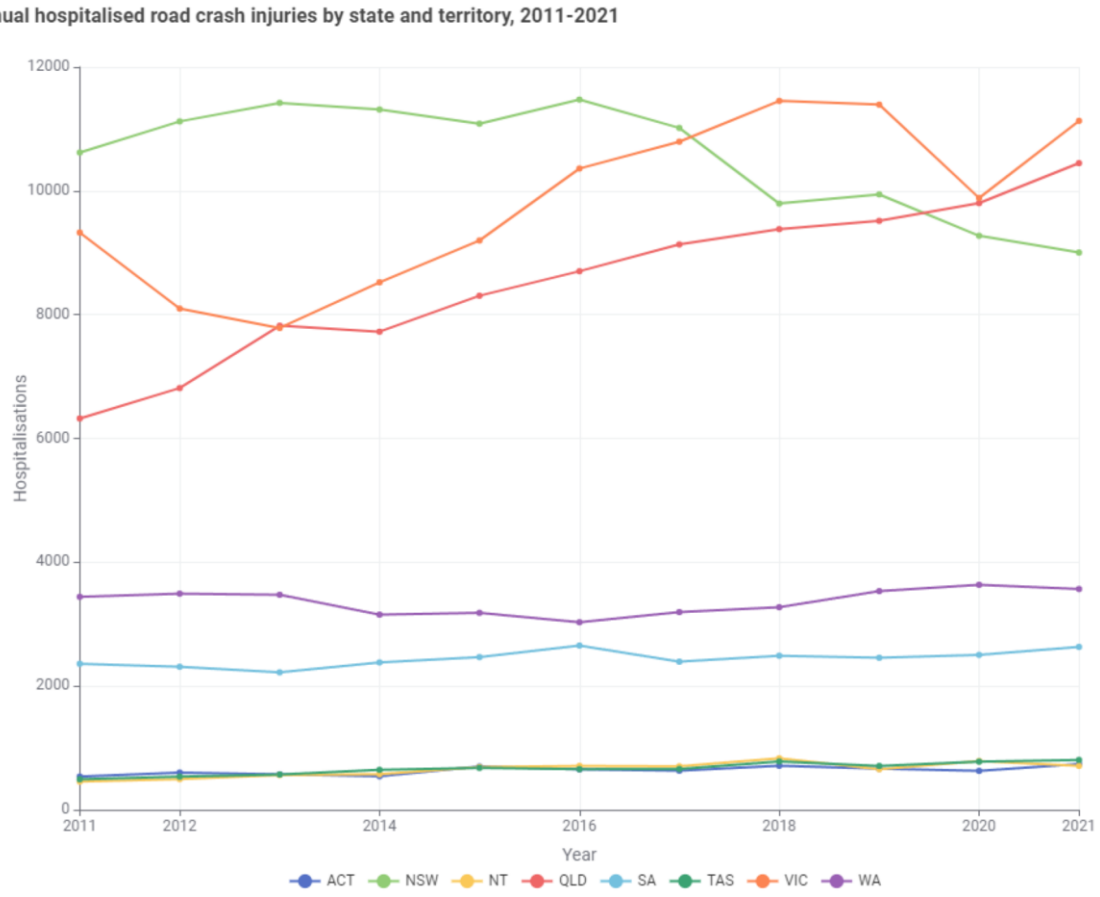


Figure 5. Annual Hospitalized Road crash injuries by state and territory, 2011-2021

Lastly, the First Nation workbook (Excel File) shows annual hospitalization by First Nations status (Figure 3). From year 2011 to 2021, it increased from 1,167 compared to Non-Indigenous Australians where it had a rise of 31,997 to 36,755. In addition, First Nations people made up 3.4% of hospitalization in 2011 and 5.6% in 2021. Lastly, when the national count dropped significantly, First Nations hospitalizations rose to 2,290 from 1,856.

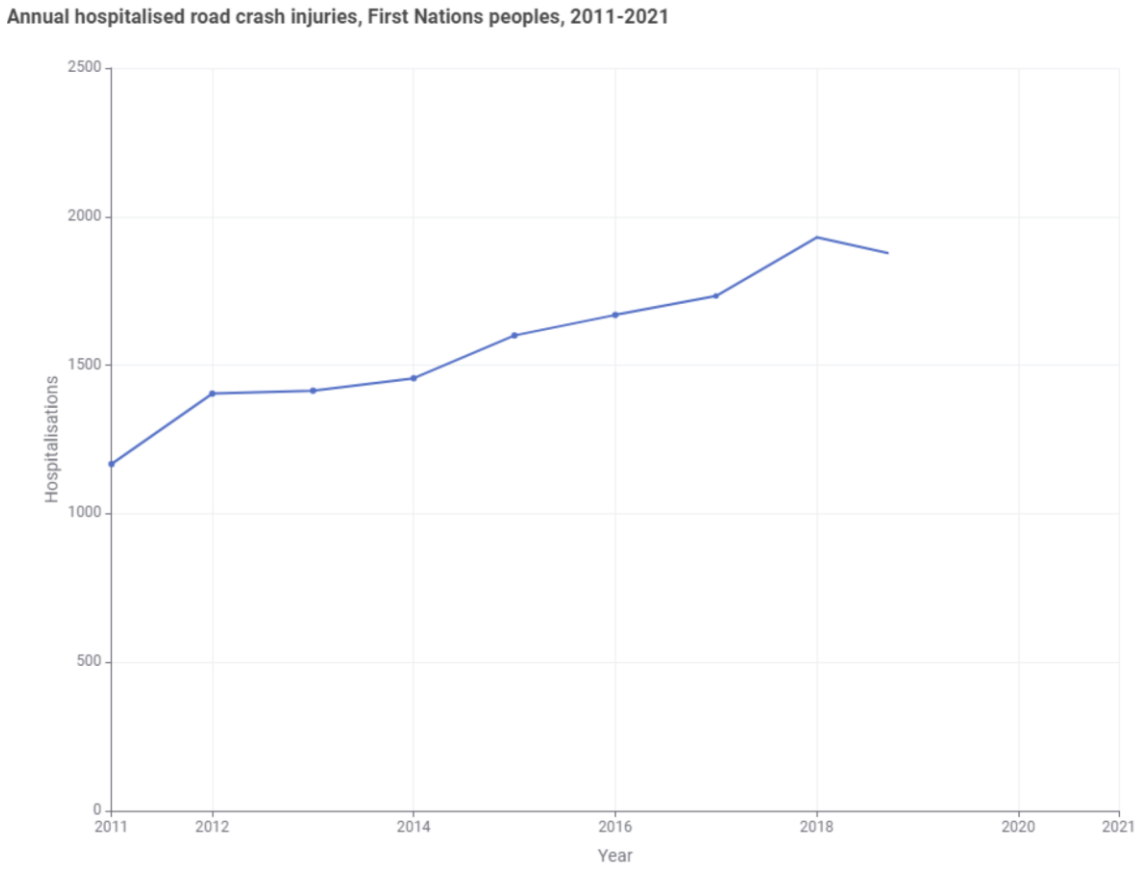


Figure 6. Annual Hospitalized Road crash injuries, First Nations peoples, 2011-2021

2.3.3 Challenges and Limitations

There are several challenges that I have identified during the initial exploration of this data.

Firstly, the raw data sheet includes hospital admissions for all injury causes in Australia, with road crash injuries combined with other injury types. Only records where “Cause of Injury” is “Traffic” match the road crash totals in Table 1, so we filtered the raw data in KNIME before using it (BITRE, 2023a).

Secondly, the original raw data contains incomplete records. For example, 6,645 of the 86,140 road crash rows have “Missing” in the ABS Remoteness Area column, and Age Group and Sex also contain “Missing” values. The Died_cases and Died_bed_days columns are empty for most rows. These issues were addressed during data processing (see Section 2.2), and the three CSV files used in this project contain no missing values.

Thirdly, the data reports absolutely count rather than rates. The higher the number of counts does not classify or interpret the data as a higher risk. This is due to the population's size. For instance, age groups that are older and states that are larger are likely to have a greater number of counts.

Visualisation Design

3.1 Website Design

3.2 Visualisation Design

3.3 Interaction Design

Iteration and Validation

4.1 Testing and Refinements

4.2 Usability Evaluation

Conclusion and Future Improvements

References

Bureau of Infrastructure and Transport Research Economics. (2023a, September 22). *Hospitalised injury*. Australian Government. https://www.bitre.gov.au/resource/road-safety/hospitalised-injury

Bureau of Infrastructure and Transport Research Economics. (2023b). *Hospitalised injury data dictionary (September 2023)*. Australian Government. https://www.bitre.gov.au/sites/default/files/documents/hospitalised-injury-data-dictionary-september-2023.pdf

Appendices

Appendix A: Generative AI Declaration

I used OpenAI's ChatGPT to help plan the data-processing steps in KNIME, troubleshoot node settings, improve workflow annotations, and draft and revise Section 2.2 of the design book. It also assisted with checking the exported CSV files and identifying inconsistencies in the report.

I implemented and executed the KNIME workflow and reviewed its outputs against the BITRE source data. The hospitalisation figures came from the BITRE datasets, rather than being generated by AI. I reviewed and edited the AI-assisted text and take responsibility for my submitted work.

Team member: Lai Man Yee

Contribution: Data processing, KNIME workflow and Section 2.2

Appendix B: Usability Evaluation Materials

Usability evaluation materials have not yet been prepared. This appendix will be updated with the testing tasks and feedback questions once the website prototype is ready for evaluation.

Appendix C: Usability Evaluation Results

Usability testing has not yet been conducted. This appendix will be updated with the evaluation results and identified usability issues after testing.